

Boruta-Driven Explainable Machine Learning for Identifying Gene Signatures in Laser-Captured Dopamine Neurons of Parkinson’s Disease

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Abstract

Background. Expression studies of the substantia nigra in Parkinson’s disease (PD) are confounded by the loss of the very cells they measure. Laser-capture microdissection (LCM) isolates dopamine neurons before sequencing, but each LCM study is small, and a panel selected in one is rarely tested anywhere else.

Methods. Four LCM datasets were merged at one profile per person (63 donors, 30 control, 33 PD; 5,622 genes measured and expressed in all four). A Random Forest on within-person gene ranks was evaluated by repeated cross-validation, leave-one-study-out and label permutation, and Boruta selected a 30-gene panel. Both models were then frozen and applied without refitting to eight public bulk nigra cohorts (158 donors after 14 shared brains were removed). We scored each donor’s dopamine-neuron content, estimated the CALB1-versus-SOX6 subtype mix from published single-nucleus markers, and repeated the whole analysis with that mix regressed out of every gene.

Results. The classifier reached an area under the ROC curve (AUC) of 0.73 (SD 0.03) across donors and 0.68 with a whole study held out (label shuffles $P = 0.005$). In bulk tissue the frozen classifier pooled 0.70 (95% CI 0.62–0.78) and the panel forest 0.84 (0.77–0.90), but eight dopamine-neuron markers alone reached 0.86. Removing neuron content left the panel at 0.76 and the classifier at 0.62. Genes lower in PD are markers of SOX6 dopamine neurons and genes higher are markers of CALB1 neurons ($P = 0.0006$; $P = 0.004$ against gene sets matched on effect size), the split that single-nucleus work reports for selective vulnerability. Over-representation analysis against the measured background returned four synapse and calcium-binding terms. When each donor’s estimated subtype mix was regressed out of every gene, out-of-fold AUC fell from 0.76 to 0.67 ($P = 0.009$) yet stayed above chance, and 23 of 30 genes kept at least half of their effect.

Conclusions. The panel carries real signal, and most of it is a shift in which dopamine neurons remain rather than a change inside surviving neurons. Reports of nigral expression signatures in PD should state how much survives an explicit composition adjustment.

Keywords: Parkinson’s disease; substantia nigra; laser-capture microdissection; external validation; cell composition; Boruta; SHAP; dopamine-neuron subtypes

1 Introduction

Parkinson’s disease is the fastest-growing neurological cause of disability worldwide (GBD 2021 Nervous System Disorders Collaborators, 2024), and its prevalence keeps rising as populations age (Ben-Shlomo et al., 2024; Bloem et al., 2021). The motor syndrome follows the loss of neuromelanin-containing dopamine neurons of the substantia nigra pars compacta (Kordower et al., 2013). Why those neurons go first is still argued over, with α -synuclein aggregation, mitochondrial and lysosomal failure and the metabolic cost of pacemaking all proposed (Morris et al., 2024; Surmeier et al., 2017). Single-nucleus sequencing sharpened the question by showing that vulnerability differs between dopamine neuron subtypes: the SOX6_AGTR1 population is depleted in PD while several CALB1 populations are relatively spared (Kamath et al., 2022).

Molecular taxonomies of midbrain dopamine neurons agree on that broad SOX6–CALB1 division (Poulin et al., 2020).

This is a problem for expression studies of nigral tissue. When neurons die the tissue that remains contains proportionally more glia, and in PD brain much of what looked like disease biology turns out to be a change in cell composition (Nido et al., 2020). Laser-capture microdissection removes part of the problem by collecting the neurons themselves before sequencing (Nichterwitz et al., 2016), and several LCM datasets of human nigral dopamine neurons are public (Zheng et al., 2010; Tiklová et al., 2021; Zaccaria et al., 2021). None of them holds more than 22 people.

Small cohorts and tens of thousands of genes are an easy way to overfit. Selecting genes on all samples before cross-validation biases the error estimate (Ambroise et al., 2002), as does tuning a model on the data used to score it (Varma et al., 2006); both remain common in genomics and in machine-learning-based science more broadly (Whalen et al., 2021; Kapoor et al., 2023). Repeated measurements from one donor are not independent (Hurlbert, 1984; Lazic, 2010), and treating pools or cells as the unit of analysis inflates false discoveries in transcriptomics (Squair et al., 2021; Zimmerman et al., 2021). The most informative test holds out whole studies (Bernau et al., 2014), because platform and batch effects can exceed the biology (Leek et al., 2010).

In earlier work we merged four LCM cohorts at one row per person and used Boruta (Kursa et al., 2010; Breiman, 2001) with SHAP (Lundberg et al., 2020) to select and explain a gene panel. Two questions were left open. First, whether the panel carries across to data the model has never seen, which for nigral expression means bulk tissue, where most public cohorts live. Second, whether what the panel measures is a change inside dopamine neurons or a change in which dopamine neurons are left to measure. Here we fix both models before touching any external data, test them in eight public bulk cohorts after removing brains shared with the discovery studies, and then ask how much of the signal survives an explicit adjustment for each donor’s estimated dopamine-neuron subtype mix.

2 Methods

2.1 Discovery cohort

All expression data came from the Gene Expression Omnibus (Barrett et al., 2012). Four LCM studies of nigral dopamine neurons were merged at one profile per person (Table 1): GSE182622, RNA sequencing of pools of neuromelanin-positive neurons (Tiklová et al., 2021); GSE20141 and GSE24378, Affymetrix HG-U133 Plus 2.0 and HG-U133 X3P profiles (Zheng et al., 2010); and GSE169755, RNA sequencing of two libraries per brain (Zaccaria et al., 2021). Replicate pools and libraries from one donor were averaged, so that 192 pools became 22 donors in GSE182622 and 12 libraries became 6 brains in GSE169755. Of 28,180 genes, 10,164 were measured on all four platforms and 5,622 were expressed in every study. Each study was standardised on its own without reference to diagnosis. A classifier trained to name each donor’s study placed every donor correctly before that step and 1% afterwards, against a 25% chance rate (Fig. 1).

Table 1: The four laser-capture discovery cohorts, one profile per person.

Dataset	Platform	Control	PD	Units merged	Source
GSE182622	RNA-seq (HiSeq 2500)	10	12	192	Tiklová et al. (Tiklová et al., 2021)
GSE20141	Affymetrix U133 Plus 2.0	8	10	18	Zheng et al. (Zheng et al., 2010)
GSE24378	Affymetrix U133 X3P	9	8	17	Zheng et al. (Zheng et al., 2010)
GSE169755	RNA-seq (HiSeq 4000)	3	3	12	Zaccaria et al. (Zaccaria et al., 2021)
Total		30	33	239	

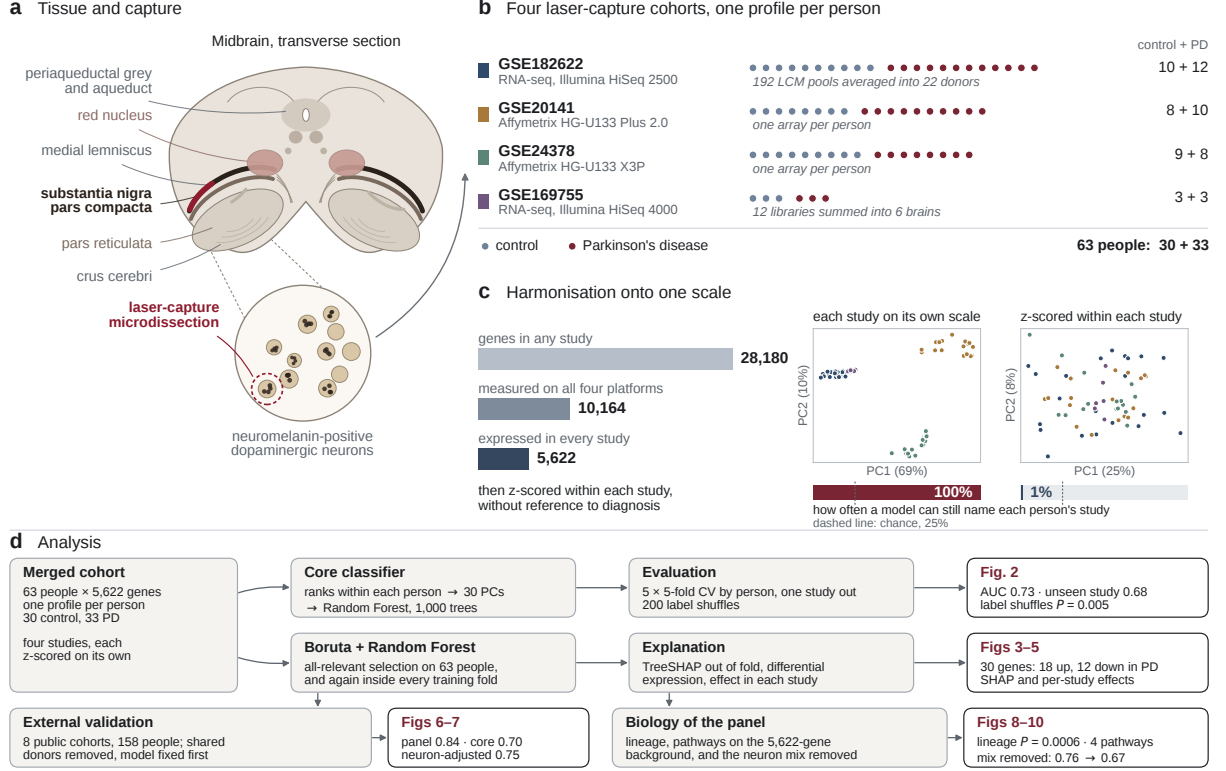


Figure 1: Study design. (a) Nigral dopamine neurons were isolated by laser-capture microdissection in each study. (b) Four cohorts, 63 people, one profile per person after replicate pools and libraries were averaged. (c) Harmonisation: 28,180 genes reduce to 5,622 measured and expressed in every study, and standardising within study removes study identity, which a classifier could otherwise recover perfectly. (d) The analysis in five stages, with the figure each one produces.

2.2 The core classifier

Three steps make up the model, and each answers a specific problem with merging studies. First, genes are ranked within each person, so a profile enters the model as the order of its genes rather than as platform-dependent intensities; a rank transform of this kind is the usual defence when arrays and sequencing are combined (Leek et al., 2010). Second, the 5,622 ranks are reduced to 30 principal components fitted on the training donors only, which keeps the dimensionality below the sample size and keeps test donors out of the projection (Ambroise et al., 2002). Third, a Random Forest (Breiman, 2001) is fitted on those components with 1,000 trees, `max_features` 0.5, one sample per leaf and balanced class weights, using scikit-learn (Pedregosa et al., 2011). A forest suits this setting because it assumes no distribution, tolerates correlated predictors, and reports an out-of-bag error from the samples each tree did not see (Breiman, 2001). Performance was estimated by five repeats of five-fold cross-validation split by person (seeds 1000–1004), by leave-one-study-out, and against 200 permutations of the diagnosis labels within each study. The model configuration was chosen inside the folds among 97 Random Forest variants, so that the reported estimate is not the maximum over the comparison (Varma et al., 2006).

2.3 The 30-gene panel

Boruta is the selection step this paper depends on, so we state it as the test it is (Kursa et al., 2010). Let $X \in \mathbb{R}^{n \times p}$ hold the $p = 5,622$ genes of $n = 63$ donors and let y be the diagnosis. At iteration t every column is copied and shuffled,

$$\tilde{X}_{\cdot j}^{(t)} = X_{\pi_j^{(t)}(\cdot), j}, \quad \pi_j^{(t)} \sim \text{Unif}(S_n), \quad (1)$$

which destroys any relation to y while keeping each gene’s own distribution; these are the shadow features. A Random Forest is fitted to $[X, \tilde{X}^{(t)}]$ and gives importances $Z_j^{(t)}$. A gene scores a hit when it beats the shadow columns of that same run,

$$h_j^{(t)} = \mathbf{1}\{Z_j^{(t)} > Q_{0.99}(Z_1^{(t)}, \dots, Z_p^{(t)})\}, \quad H_j = \sum_{t=1}^T h_j^{(t)}, \quad (2)$$

and under the null that the gene carries nothing, $H_j \sim \text{Binomial}(T, \frac{1}{2})$. Genes are confirmed or rejected by a two-sided binomial test on H_j with a Bonferroni correction over the p genes; rejected genes are dropped from later iterations, and whatever is still undecided when t reaches T stays tentative. Using $Q_{0.99}$ of the shadow importances instead of their maximum is the stricter variant, which suits a 63-donor cohort.

Boruta is all-relevant rather than minimal-optimal: it keeps every gene that carries signal instead of the smallest set that predicts, which is what one wants when the aim is interpretation (Kursa et al., 2010; Degenhardt et al., 2017). We used $T = 100$ iterations and a 500-tree forest, run on all 63 donors and again inside every training fold, so Table 3 reports how often each gene is reselected rather than assuming the selection is stable (Degenhardt et al., 2017). The panel holds 30 genes, 18 higher and 12 lower in PD.

TreeSHAP values (Lundberg et al., 2020) were computed out of fold. Differential expression used the Wilcoxon rank-sum test (Mann et al., 1947) with Benjamini-Hochberg control (Benjamini et al., 1995), and per-study effects were combined as Hedges’ g (Hedges, 1981) in a random-effects meta-analysis (DerSimonian et al., 1986).

2.4 Model locking and external cohorts

Both models, the core classifier and a forest restricted to the 30 panel genes, were fitted on all 63 discovery donors and stored before any external cohort was downloaded. Nothing was refitted, re-selected or re-tuned afterwards. Eight public bulk substantia nigra cohorts were then scored: GSE7621 (Lesnick et al., 2007), GSE8397 (Moran et al., 2006), GSE20163 and GSE20164 (Zheng et al., 2010), GSE20292 (Zhang et al., 2005), GSE49036 (Dijkstra et al., 2015), GSE114517 (Simchovitz et al., 2020) and GSE168496 (Tranchevent et al., 2023). Probe sets were mapped to gene symbols through the g:Profiler conversion service (Kolberg et al., 2023); HG-U133A arrays were mapped through the Plus 2.0 namespace, which recovers 4,584 of the 5,622 discovery genes. Each cohort was standardised within itself and then ranked within each person, exactly as in discovery.

Donor identifiers were compared with the discovery studies. GSE20292 shares 12 brains with GSE20141 and GSE20163 shares 2, identified by matching donor numbers; those donors were removed, leaving 158 people (72 control, 86 PD). Because shared brain banks can leak in ways identifiers do not show, every cohort was also scored by strictly independent models retrained without any discovery study from the same bank. Significance came from 10,000 permutations of diagnosis within cohorts; confidence intervals from 4,000 donor bootstraps (Efron, 1979). Cohort AUCs were pooled with weights equal to the number of PD–control pairs, and ROC curves were averaged vertically with the same weights so that the drawn curve has the pooled area (Hanley et al., 1982).

2.5 Dopamine-neuron content and subtype mix

Two adjustments were made, and they answer different questions. The first asks whether a score merely tracks how much dopamine-neuron material a bulk sample contains: eight dopamine-neuron markers were averaged into a neuron score, and each model score was residualised on it within each cohort. The second asks which dopamine neurons remain. Marker z scores for every dopamine-neuron subtype were taken from Supplementary Table 8 of Kamath et al. (2022); a lineage score was defined for each gene as its mean marker strength over the six CALB1 subtypes

minus its mean over the four SOX6 subtypes. The 50 strongest markers on each side, with panel genes excluded, gave a per-donor composition score, standardised within study. That score was then regressed out of every one of the 5,622 genes, and the whole discovery pipeline was run again on the adjusted data with the same folds and the same model. Sensitivity runs used 25, 100 and 200 markers, SOX6_AGTR1 markers alone, and composition together with overall dopamine purity.

2.6 Pathway analysis

Over-representation was tested with g:GOST (Kolberg et al., 2023) using the 5,622 measured genes as the background and Benjamini-Hochberg correction across all 3,547 terms (Benjamini et al., 1995), against GO, Reactome (Milacic et al., 2023), KEGG (Kanehisa et al., 2022) and WikiPathways (Agrawal et al., 2023). Because a gene list of this size can return terms by chance, the same settings were applied to 100 random gene lists of each size drawn from the same background, and to lists matched on differential-expression rank; the calibrated P value is the share of those lists reaching a smaller adjusted P than the panel. A ranking-based analysis was also run for completeness: gene set enrichment analysis (Subramanian et al., 2005) over all 5,622 genes against 1,967 sets from Hallmark (Liberzon et al., 2015), Reactome, KEGG and GO, with the null built by shuffling donor diagnoses within study rather than genes, and single-sample scores per donor (Barbie et al., 2009). Gene set definitions came from Enrichr’s libraries (Kuleshov et al., 2016).

2.7 Gene-level annotation

Table 3 collects what is known about each panel gene, and every cell in it is either measured here or taken from a named public resource at a stated release. Protein names and functions are from UniProt (The UniProt Consortium, 2024), with the accession given so each statement can be checked. Association scores with Parkinson disease, the tractability evidence and the count of drug or clinical candidates are from the Open Targets Platform (Buniello et al., 2024). Dopamine-neuron subtype identity is the lineage score described above (Kamath et al., 2022).

Published links to the disease were found with one protocol applied to every gene, so that the absence of a link is as checkable as its presence. We searched Europe PMC (Rosonovski et al., 2023) for the gene symbol, its approved name and every HGNC alias, in the title or abstract, together with Parkinson, parkinsonism, Lewy, synuclein, LRRK2, PINK1, parkin, glucocerebrosidase, dopaminergic neuron, substantia nigra, MPTP, rotenone, 6-hydroxydopamine or paraquat; for open-access papers we also read the full text sentence by sentence. A paper was kept only when one sentence reports a finding about the gene itself in that setting. Reviews, acronym collisions (*FUS* returns focused ultrasound, *SCAI* a cardiology society) and retracted papers were excluded, and every citation was checked against Crossref for retractions and corrections. The complete search, with each hit and its disposition, is Supplementary Table S1.

2.8 Reproducibility

Every analysis ran in public Kaggle notebooks, each writing the tables the figures read; no figure recomputes a result. Random seeds are fixed and reported in the notebooks.

3 Results

3.1 One profile per person, and a classifier that works at donor level

The merged cohort holds 63 people from four studies (Table 1), one profile each, on 5,622 genes. Standardising within study removed study identity almost completely (Fig. 1c).

On these donors the classifier reached an AUC of 0.725 (SD 0.032) and an accuracy of 0.692 across 25 person-level folds, with sensitivity 0.56 and specificity 0.83 (Fig. 2). Holding out a whole study gave 0.678, ranging from 1.00 in the six-brain GSE169755 to 0.55 in GSE20141. Against 200 label permutations the observed AUC of 0.707 sat above a null centred on 0.498 ($P = 0.005$), and choosing the model inside the folds rather than once on all data gave 0.741, so the estimate is not an artefact of the comparison (Varma et al., 2006). A forest on all 5,622 genes without the rank and component steps reached 0.66.

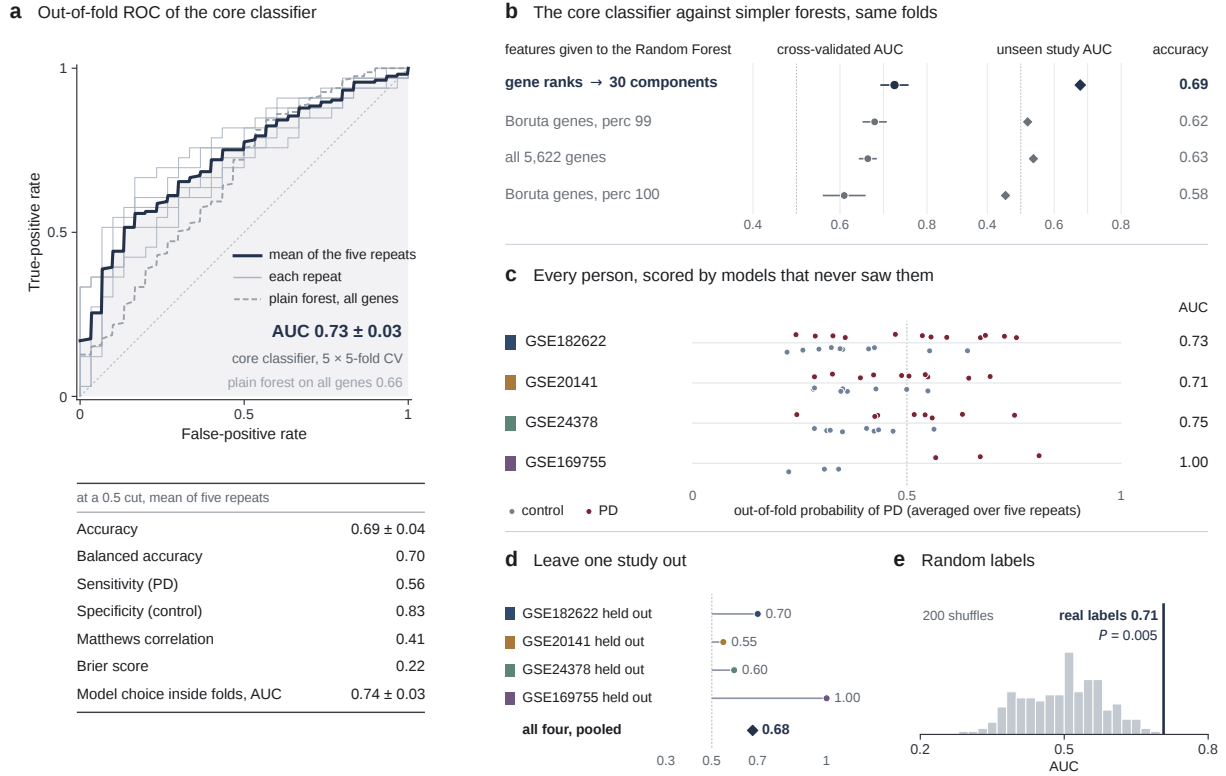


Figure 2: The core classifier. (a) Out-of-fold ROC over 25 person-level folds. (b) The same folds for simpler forests. (c) Every donor’s out-of-fold score. (d) Leave-one-study-out. (e) 200 permutations of the diagnosis labels within study.

3.2 What the model uses

Boruta confirmed 30 genes, 18 higher and 12 lower in PD. *RBM4B*, *PLXNC1* and *GPNMB* lead the SHAP ranking and were kept in 100%, 96% and 92% of training folds; single genes reach AUCs of 0.63–0.78 on their own (Fig. 3, Fig. 4).

Twenty-two of the 30 are also nominally differentially expressed at $P < 0.01$, an overlap 79 times larger than chance would give (hypergeometric $P = 6 \times 10^{-41}$; Fig. 5). No gene survives a 5% false-discovery rate in the 63 donors, which is the expected outcome at this sample size rather than evidence of absence (Benjamini et al., 1995).

3.3 The frozen models in eight bulk cohorts

Both models were fixed before the external data were touched. Applied without refitting to 158 donors in eight bulk nigra cohorts, the core classifier pooled an AUC of 0.70 (95% CI 0.62–0.78) and the panel forest 0.84 (0.77–0.90), both far from chance under within-cohort label permutation ($P < 0.001$; Fig. 6a, Table 2). At a cut-off fixed before testing the panel called 72% of donors correctly, with sensitivity 0.76 and specificity 0.68 (Fig. 7b). Per-cohort AUCs of the panel ranged from 0.43 in the 15-donor GSE20163 to 0.98 in GSE168496 (Fig. 7a). Retraining without any

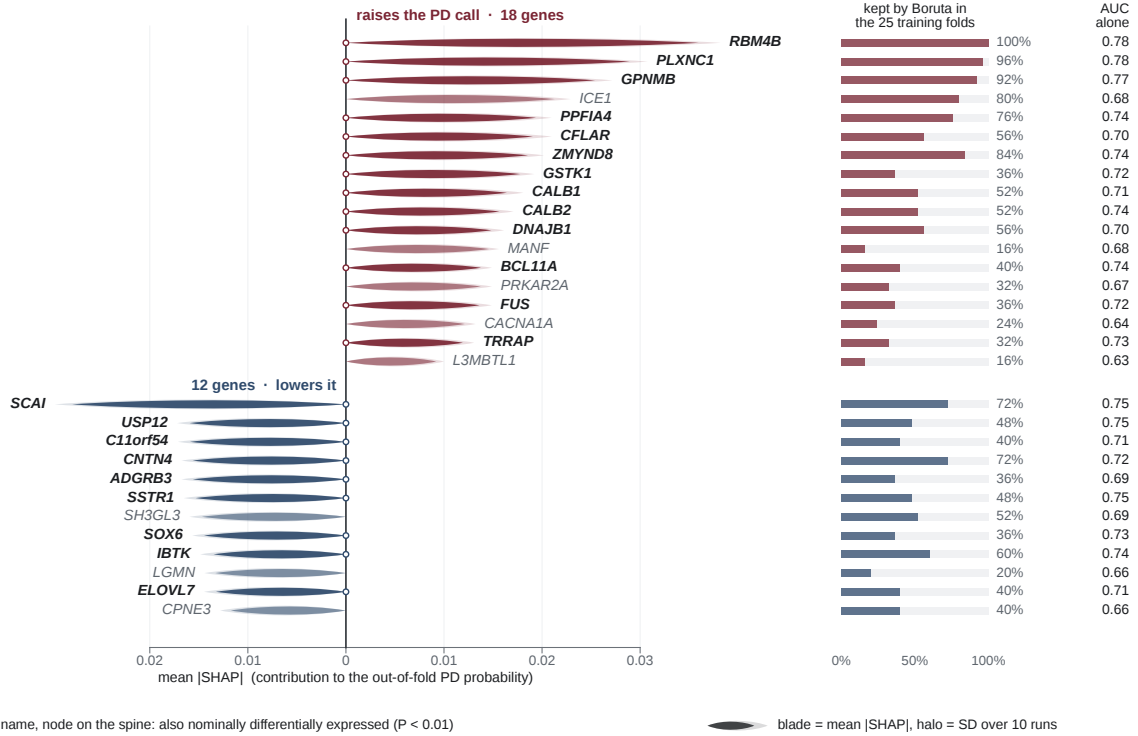


Figure 3: The 30 Boruta genes ranked by mean absolute SHAP contribution to the out-of-fold PD probability, split by direction, with the share of training folds that reselected each gene and the AUC of each gene alone. Bold names are also nominally differentially expressed.

discovery study from the same brain bank changed little: 0.80 for the panel and 0.73 for the classifier. Dropping GSE7621, the cohort most similar to the discovery arrays, left the panel at 0.82.

Chance is the wrong comparison here. Eight dopamine-neuron markers, averaged into a single score with no model behind them, reached 0.86 in the same donors, better than either model. Removing that neuron component from the scores within each cohort dropped the panel to 0.76 and the classifier to 0.62 (Table 2). Gene by gene, 19 of 30 effects had the same sign in bulk as in the laser-captured neurons, which is not more than chance ($P = 0.10$); after the neuron score is removed from every gene, 23 of 30 agree ($P = 0.003$, Spearman $\rho = 0.72$; Fig. 6c). So the panel does say something about bulk nigra beyond counting neurons. Counting neurons still does the job at least as well.

Table 2: Pooled performance in the eight external bulk-tissue cohorts (158 people, 72 control, 86 PD). *Strictly independent* models were retrained without any discovery study from the same brain bank. *Neuron score removed* residualises each donor’s score on eight dopamine-neuron markers. Confidence intervals are 4,000 donor bootstraps.

Model	AUC [95% CI]	Neuron score removed	Label shuffles
Core classifier, fixed	0.70 [0.62–0.78]	0.62 [0.53–0.71]	< 0.001
Core classifier, strictly independent	0.73 [0.65–0.81]	0.67 [0.58–0.76]	–
30-gene panel forest	0.84 [0.77–0.90]	0.75 [0.67–0.83]	< 0.001
30-gene panel, strictly independent	0.80 [0.73–0.86]	0.70 [0.61–0.78]	–
Dopamine-neuron markers alone	0.86 [0.79–0.92]	–	–



Figure 4: SHAP values of every donor for every panel gene, coloured by that donor's expression rank.

3.4 The panel tracks a dopamine-neuron lineage

Three quarters of the panel genes appear in the Kamath single-nucleus marker table (Kamath et al., 2022), and they fall apart cleanly by direction. Genes lower in PD are markers of *SOX6* dopamine neurons; genes higher are markers of *CALB1* neurons (difference in lineage score $P = 0.0006$; each side also differs from the measured background, $P = 0.003$ and $P = 0.002$; Fig. 8). Because effect size and lineage could be correlated through expression level, the same test was run against gene sets matched on differential expression rank, and the split survived ($P = 0.004$). *SOX6* itself is in the panel, lower in PD, as is *ADGRB3*; *CALB1* and *CALB2* are higher. This is the direction single-nucleus work reports for selective vulnerability (Kamath et al., 2022), recovered here from four bulk-processed LCM studies that never saw a single-cell library.

3.5 Pathways, with an honest background

Using the 5,622 measured genes as the background and correcting across all 3,547 terms, the 30-gene panel returns four significant terms: calcium ion binding involved in regulation of presynaptic cytosolic calcium (FDR 0.005), excitatory synapse (0.016), a GO term labelled after cerebellar climbing-fibre synapses that covers the same excitatory synaptic genes (0.018), and regulation of synapse pruning (0.028; Fig. 9). Four terms is more than most random lists of the same size return: 32% of random lists return at least one, 10% return four or more, and the panel's strongest term beats 98% of them. The enrichment is real and modest. The terms rest on five genes: *CALB1*, *CALB2*, *ADGRB3*, *PLXNC1* and *PPFIA4*. Four of those five are also lineage markers, so this is close to the result of the previous section in different words.

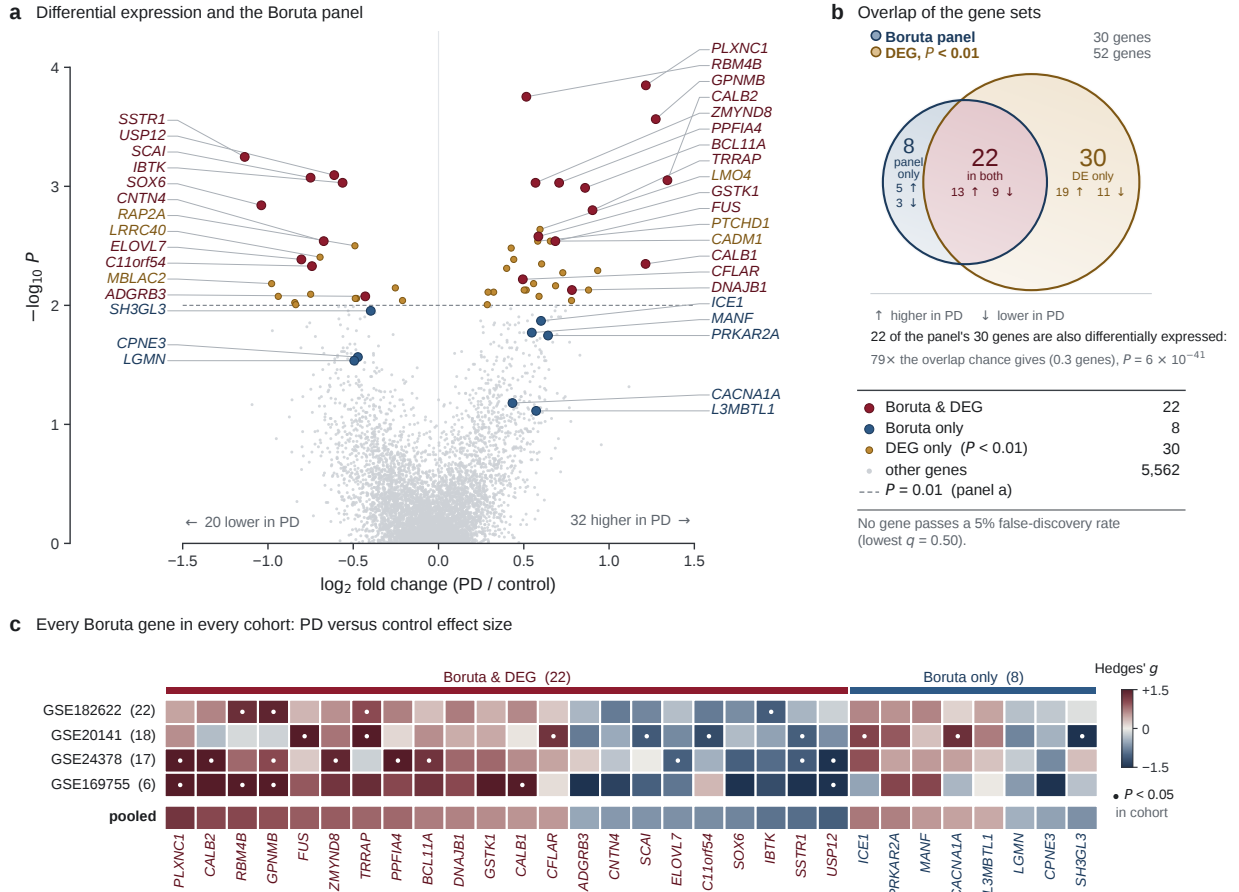


Figure 5: Differential expression and the panel. (a) Volcano of all 5,622 genes. (b) Overlap of the Boruta panel with genes at $P < 0.01$: 22 of 30, with the up and down split of each region; the overlap is 79 times what chance gives. (c) Effect of every panel gene in every discovery cohort.

A ranking-based analysis adds nothing that survives a strict null. Across 1,967 gene sets, none reaches significance when the null is built by shuffling donor diagnoses within study; 20 pass the conventional gene-permutation FDR of 0.25, of which 12 carry a panel gene in the leading edge, and single-sample scores separate PD from control at $P < 0.05$ for 147 sets but none after correction. We report those as exploratory, not as findings (Subramanian et al., 2005).

3.6 How much of it is the neuron mix

Each donor's dopamine-neuron subtype mix was estimated from the Kamath markers, with panel genes excluded. PD donors are more CALB1-like than controls (AUC 0.73, Mann-Whitney $P = 0.0003$), and the classifier's out-of-fold score follows that mix ($\rho = 0.51$ overall, 0.60 among controls alone), which is the correlation one expects if the model is partly reading composition (Fig. 10a,b).

Regressing the mix out of every gene and re-running the same folds with the same model is the test that separates the two explanations. Out-of-fold AUC falls from 0.765 to 0.674, a drop that is well outside the fold-to-fold variation, and the adjusted model still beats its own label permutations ($P = 0.009$; Fig. 10c). Adding the classifier score to a model built on composition alone improves it (leave-one-out AUC 0.68 to 0.72, likelihood-ratio $P = 0.006$). Sensitivity runs with 25, 100 or 200 markers, with SOX6_AGTR1 markers alone, or with dopamine purity added, land between 0.64 and 0.76. Gene by gene, 23 of 30 keep at least half of their effect, with a median of 68% kept (Fig. 10d, Table 3).

So the signal is both things at once. A large part of what the classifier reads is which dopamine

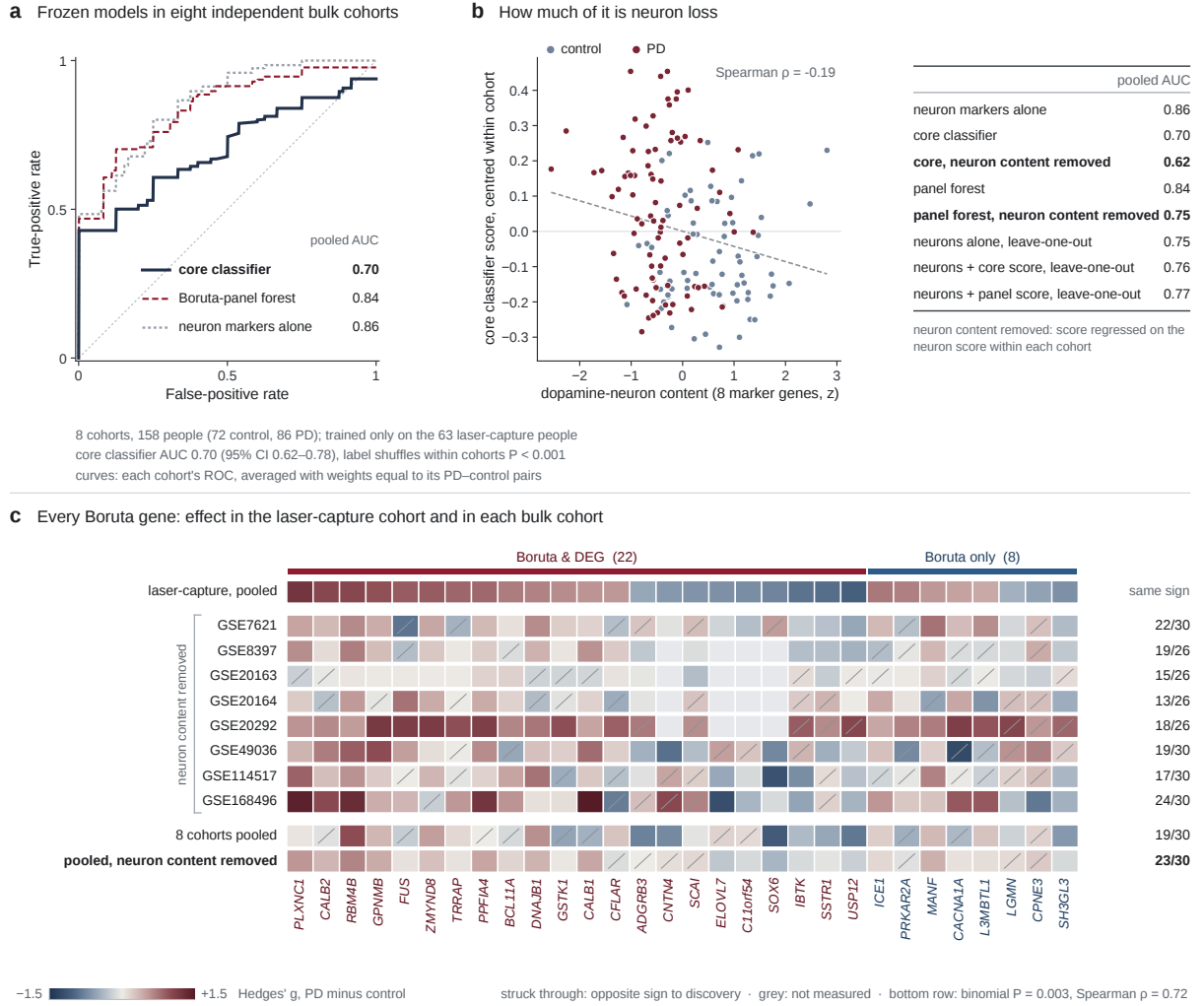


Figure 6: External validation in eight bulk cohorts, 158 donors, both models frozen beforehand. (a) Weighted average ROC of the classifier, the panel forest and a dopamine-neuron marker score. (b) Classifier score against neuron content, with pooled AUCs after that content is removed. (c) Each panel gene's effect in discovery and in bulk tissue, before and after the neuron adjustment.

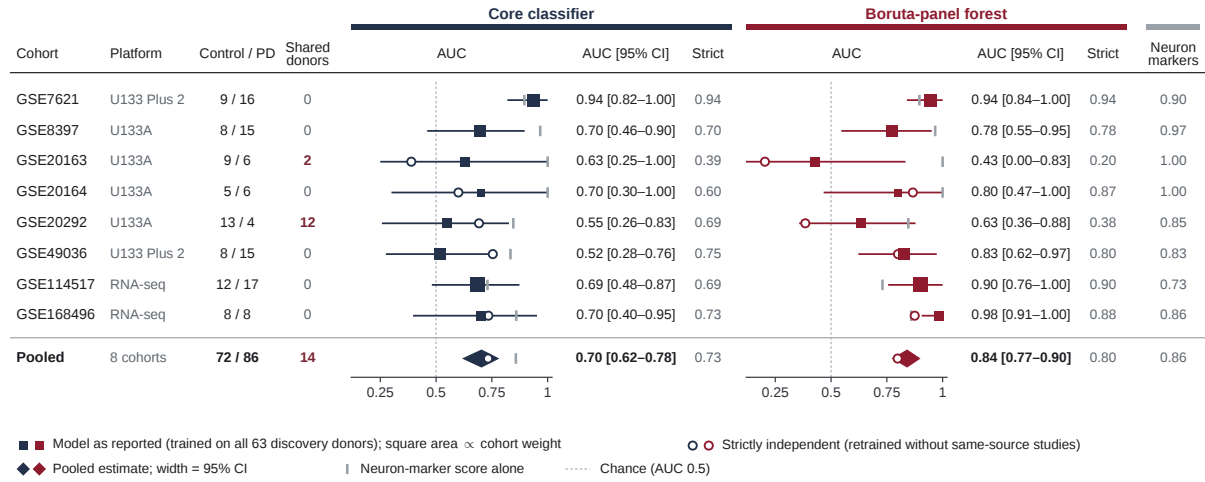
neurons survive; a smaller part is a difference in the neurons that remain.

3.7 The genes themselves

Table 3 gives all 30 genes: the four measurements above, the dopamine-neuron subtype each one marks, its Open Targets association with Parkinson disease, what is tractable about it, and any published study tying it to the disease or to dopamine neurons.

Thirteen of the 30 carry such a study. Two of them, *GPNMB* and *ELOVL7*, sit at genetic risk loci (Open Targets genetic association 0.78 and 0.51) (Buniello et al., 2024; Blauwendraat et al., 2020). *SOX6* and *CALB1* are the markers of the vulnerable and the spared dopamine populations (Kamath et al., 2022; Damier et al., 1999), and *CALB2* is the third calcium-binding protein in that group: in human PD midbrain its neurons were selectively preserved (Mouatt-Prigent et al., 1994). *LGMN* encodes legumain, which cleaves α -synuclein at N103 and also truncates *SOX6* in vulnerable neurons (Zhang et al., 2017; Nie et al., 2024). *DNAJB1* clears α -synuclein through the Hsp70 system (Huang et al., 2025), *MANF* restores nigrostriatal function after a 6-hydroxydopamine lesion (Voutilainen et al., 2009), and *BCL11A* marks a nigral subset that is especially vulnerable to α -synuclein overexpression (Tolve et al., 2021). That a wrapper algorithm with no knowledge of any of this recovered thirteen such genes is the strongest evidence that the

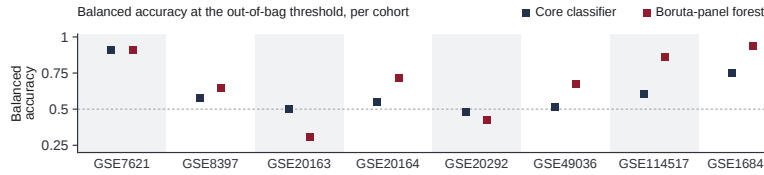
a Discrimination in each external cohort



b Classification at pre-specified thresholds

	AUC [95% CI]	Threshold 0.5			Out-of-bag threshold		
		Accuracy	Sensitivity	Specificity	Accuracy	Sensitivity	Specificity
■ Core classifier	0.70 [0.62–0.78]	0.65	0.56	0.75	0.62	0.64	0.60
○ strictly independent	0.73 [0.65–0.81]	0.58	0.59	0.56	0.57	0.63	0.50
■ Boruta-panel forest	0.84 [0.77–0.90]	0.72	0.76	0.68	0.72	0.71	0.74
○ strictly independent	0.80 [0.73–0.86]	0.70	0.74	0.65	0.73	0.66	0.82

Pooled over 8 cohorts (158 donors). Out-of-bag thresholds set on the 63 discovery donors: core 0.44, panel 0.54.



c Donor overlap with discovery

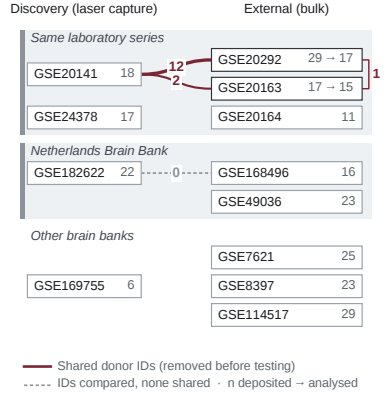


Figure 7: The frozen models cohort by cohort. (a) AUC of each cohort with bootstrap intervals and the pooled estimate. (b) Calls at cut-offs fixed before testing. (c) Brains shared with the discovery studies, found by donor identifier and removed.

selection is not noise.

The other seventeen have no published tie to the disease, which is what makes them candidates rather than confirmations. Which of them to follow depends on the composition adjustment rather than on the literature: *FUS* and *DNAJB1* keep 90% and 78% of their effect once the neuron mix is removed, while *CNTN4* and *CALB2* keep about a third, which is what a subtype marker looks like.

4 Discussion

A classifier trained on 63 laser-captured donors transfers to bulk nigra it has never seen: 0.70 for the model, 0.84 for the 30-gene panel. The panel's directions match the SOX6–CALB1 split that single-nucleus sequencing reports for selective vulnerability (Kamath et al., 2022). When each donor's estimated subtype mix is regressed out of every gene, the classifier loses about half of its margin over chance and keeps the rest.

The external result has to be read against the neuron-marker comparison. Eight markers with no model behind them reach 0.86 in the same cohorts, above both trained models, and removing that component costs the panel 0.08 of AUC and the classifier 0.08 more. Composition explaining much of what looks like differential expression is the rule in nigral tissue, not the exception (Nido et al., 2020). The panel is not reducible to it: after the neuron score is removed, 23 of 30 gene

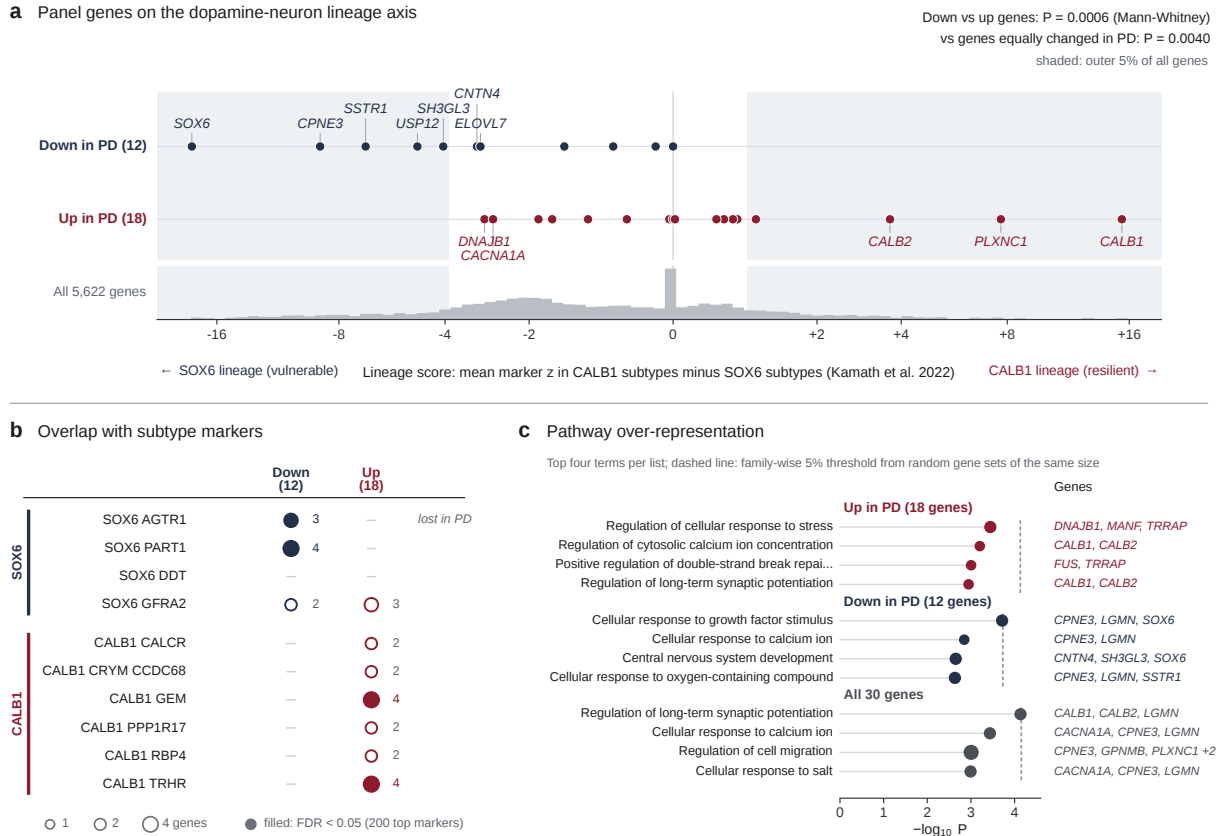


Figure 8: The panel against dopamine-neuron subtypes (Kamath et al., 2022). (a) PD effect and lineage score of every measured gene, with the panel marked. (b) Marker overlap by subtype. (c) Marker strength of each panel gene in each subtype: genes lower in PD are SOX6 markers, genes higher are CALB1 markers.

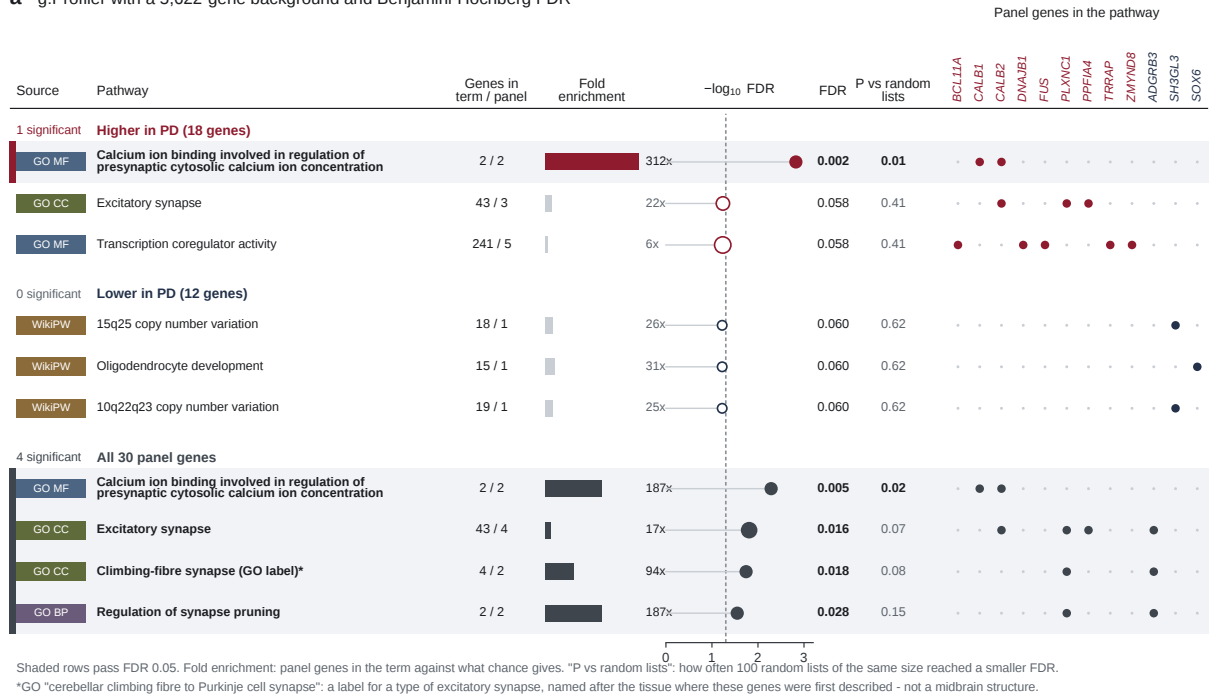
effects still agree in direction between laser-captured neurons and bulk tissue ($P = 0.003$), which a pure composition marker would not do. But any claim that a nigral expression signature is diagnostic has to clear this bar first, and most published signatures have never been asked to.

We did not expect the subtype result to hold. Boruta (Kursa et al., 2010) knows nothing about cell types, the four studies predate the single-nucleus atlas, and the genes still sort by lineage at $P = 0.0006$, and $P = 0.004$ against sets matched on effect size. That an unsupervised selection recovered an axis established independently (Kamath et al., 2022; Poulin et al., 2020) is reassuring about the method. It is also the least comfortable finding here, because a panel that reports which dopamine neurons are left is measuring a consequence of the disease rather than a mechanism inside the cells.

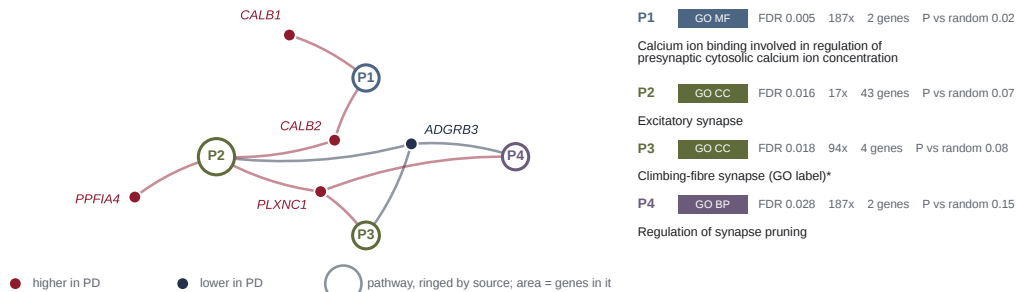
What survives the adjustment is a smaller signal, not an absent one. *FUS*, *DNAJB1*, *TRRAP* and *ZMYND8* keep the largest share of their effect, and they are chaperone, RNA-binding and chromatin genes rather than subtype markers. *DNAJB1* belongs to the HSP40 chaperone family, whose members differ in how well they suppress seeded α -synuclein aggregation (Deshayes et al., 2019), and *FUS* is an RNA-binding protein central to work on RNA processing in neurodegeneration (Lagier-Tourenne et al., 2010). Neither is a PD mechanism on this evidence. They are where we would look next, because their signal does not dissolve when composition is removed.

Cause and consequence stay entangled here. All the tissue is post-mortem, and PD donors have already lost most of the neurons the study is about (Kordower et al., 2013). The adjustment itself may also remove real biology along with the mix, since subtype markers and disease response are not independent, and regressing out one costs some of the other. That is why the sensitivity runs matter: the result holds between 25 and 200 markers, with SOX6_AGTR1 markers alone, and with dopamine purity added.

a g:Profiler with a 5,622-gene background and Benjamini-Hochberg FDR



b Every significant pathway and the panel genes that carry it



Four pathways rest on five genes, three of them shared. Those genes are largely markers of which dopamine neurons survive, so part of this signal is the neuron mix rather than a pathway (Figure 10).

c The same settings on 100 random gene lists of the same size

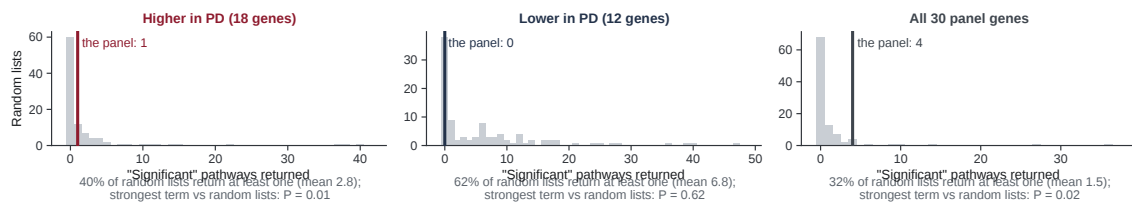


Figure 9: Over-representation with the 5,622 measured genes as background and Benjamini-Hochberg correction. (a) Terms for the lower, higher and full panel. (b) The four significant terms and the five genes that carry them. (c) The same settings applied to 100 random gene lists of each size.

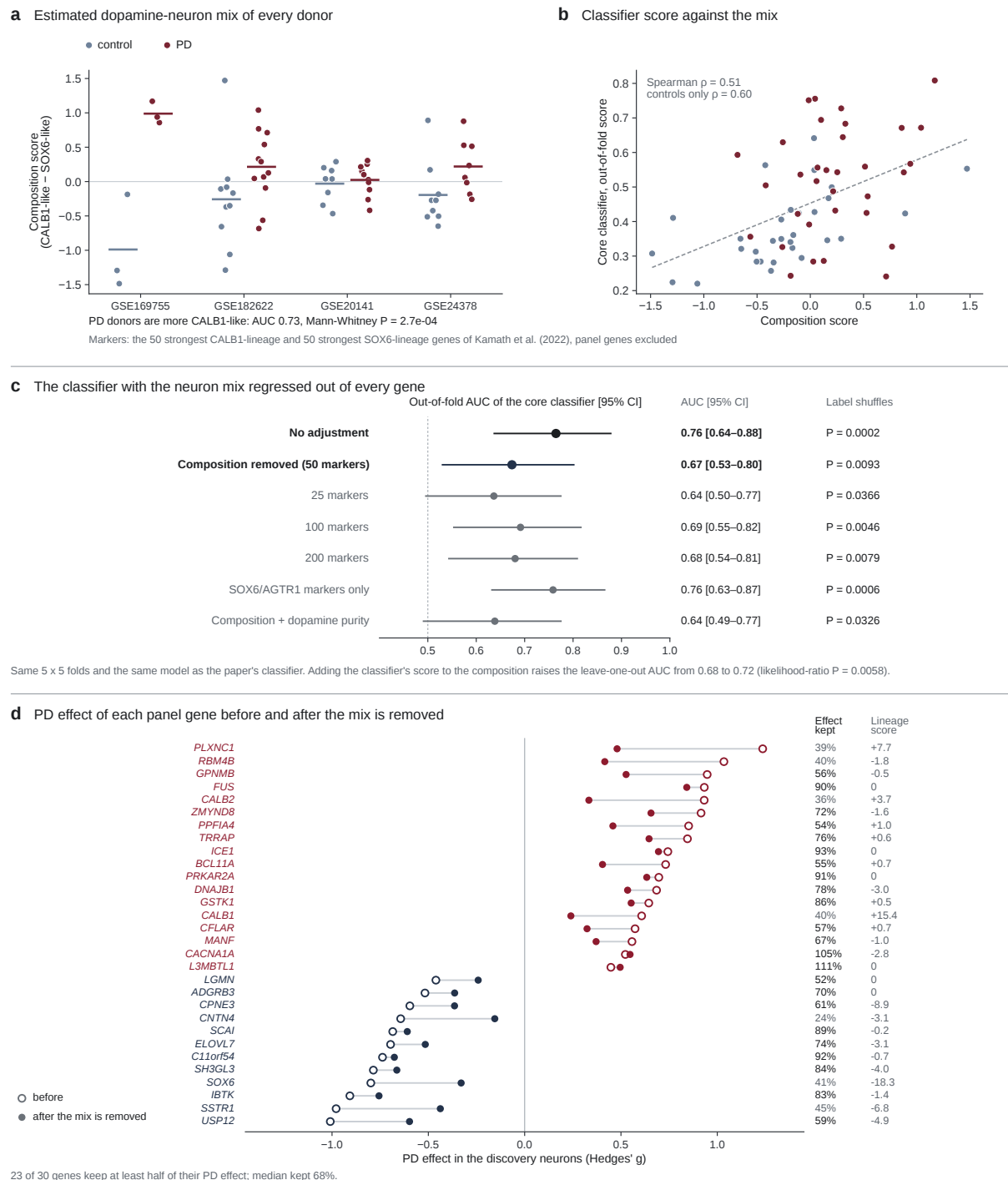


Figure 10: How much of the signal is the neuron mix. (a) Estimated CALB1-versus-SOX6 mix of every donor. (b) The classifier's out-of-fold score against that mix. (c) Out-of-fold AUC with the mix regressed out of every gene, across marker counts and variants. (d) Each panel gene's effect before and after the adjustment.

Limitations

The discovery cohort is 63 people, which is large for laser-capture nigra and small for machine learning; no gene passes a genome-wide false-discovery threshold, and the panel should be read as a stable selection, not a list of established PD genes. The external cohorts are bulk tissue, so the transfer is across both study and tissue type, and cohort sizes run from 11 to 29 donors, which makes individual cohort AUCs unstable. Donor overlap could only be excluded where identifiers allowed it; the strictly independent models are the guard against what identifiers missed. Subtype mixtures are estimated from marker averages, not measured, and the Kamath markers come from a different cohort and technology (Kamath et al., 2022). Finally, clinical detail is thin in the public metadata: we could not test disease duration, medication or Braak stage systematically, though the early-stage cohort GSE49036 (Dijkstra et al., 2015) scored no differently from the rest.

Data and code availability

Every dataset used here is public in the Gene Expression Omnibus (Barrett et al., 2012). The discovery cohort is four laser-capture studies of nigral dopamine neurons: GSE182622 (Tiklová et al., 2021), GSE20141 and GSE24378 (Zheng et al., 2010), and GSE169755 (Zaccaria et al., 2021). The models were then tested in eight bulk substantia nigra cohorts: GSE7621 (Lesnick et al., 2007), GSE8397 (Moran et al., 2006), GSE20163 and GSE20164 (Zheng et al., 2010), GSE20292 (Zhang et al., 2005), GSE49036 (Dijkstra et al., 2015), GSE114517 (Simchovitz et al., 2020) and GSE168496 (Tranchevent et al., 2023). Dopamine-neuron subtype markers are Supplementary Table 8 of Kamath et al. (2022). Gene annotation came from UniProt (The UniProt Consortium, 2024), target information from the Open Targets Platform (Buniello et al., 2024), and the literature search from Europe PMC (Rosonovski et al., 2023).

All code is public at <https://github.com/Alisareminia/pd-lcm-boruta>: the seventeen notebooks as they were run, the scripts that generate them, the tables behind every figure, and the figures themselves. Each notebook reads the outputs of the ones before it, so any figure can be redrawn without repeating the analysis. The record of the literature search, with every hit and whether it was kept, is Supplementary Table S1.

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Table 3: The 30 Boruta genes, ordered by SHAP rank. *Type and main role:* the UniProt protein name and its function, summarised in one sentence (The UniProt Consortium, 2024); bold names are pathway terms that pass Benjamini-Hochberg correction against the measured background. *This study:* four measurements, each bar scaled to the largest value among the 30 genes. *effect* is Hedges’ g between 33 PD and 30 control donors; *folds* is the share of training folds in which Boruta reselected the gene; *bulk* is the inverse-variance pooled g in eight external bulk cohorts after dopamine-neuron content is regressed out (grey bar: sign opposite to discovery); *kept* is g after each donor’s CALB1/SOX6 composition is regressed out of every gene, the bar showing the share of the original effect that remains. Each q is a Benjamini-Hochberg value across the 30 genes within its own family of tests: the Wilcoxon rank-sum test for the discovery effect, and z tests on the pooled or adjusted effect against its standard error for the other two. *DA-neuron subtype:* CALB1 minus SOX6 lineage score and the subtype in which the gene is the strongest marker, from single-nucleus profiling of human nigra (Kamath et al., 2022). *Open Targets:* the overall association score with Parkinson disease, and beneath it the genetic-association component, which integrates GWAS and rare-variant evidence, where it is non-zero (Buniello et al., 2024). *Druggability:* Open Targets tractability evidence by modality and the number of drug or clinical candidates; for each modality only the strongest tier in Open Targets’ own hierarchy is shown. *Published link:* primary studies tying the gene to Parkinson’s disease or dopamine-neuron biology, study type first, and any measurement of the gene in patients’ blood or cerebrospinal fluid. They were found with one protocol for every gene (Methods): Europe PMC (Rosonovski et al., 2023), searched for the gene symbol, its protein name and every HGNC alias together with Parkinson’s-disease and dopamine-neuron terms, in titles, abstracts and open-access full text; each hit was read and kept only when it reports a finding about the gene itself, and reviews, acronym collisions and retracted papers were excluded. The whole search, every hit and whether it was kept, is Supplementary Table S1. Empty cells mean the resource holds no entry. The literature search ran on 2026-09-21; Open Targets release 26.06 was accessed on 2026-09-21; $\uparrow$ higher and $\downarrow$ lower in Parkinson’s disease; loci are GRCh38.

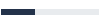
Gene	Type and main role	This study	DA-neuron subtype	Open Targets	Druggability	Published link to Parkinson’s disease or dopamine neurons
<i>RBM4B</i> $\uparrow$ 11q13.2 SHAP rank 1	RNA-binding protein 4B RNA-binding protein that activates translation of the clock gene PER1. (The UniProt Consortium, 2024) [Q9BQ04]	effect  $g +1.03, q 0.003$ folds  100% of folds bulk  $g +0.66, q 0.004$ kept  $g +0.42, q 0.154$	-1.8 SOX6 DDT	0.000		
<i>PLXNC1</i> $\uparrow$ 12q22 SHAP rank 2	Plexin-C1 Receptor for semaphorin 7A; its signalling rearranges the cytoskeleton. (The UniProt Consortium, 2024) [O60486] excitatory synapse, climbing-fibre synapse, synapse pruning	effect  $g +1.23, q 0.003$ folds  96% of folds bulk  $g +0.56, q 0.015$ kept  $g +0.48, q 0.107$	+7.7 CALB1 CALCR	0.001	<i>small molecule:</i> high-quality pocket <i>antibody:</i> surface/secreted (high confidence)	Mouse development: semaphorin 7A–PLXNC1 signalling separates the nigrostriatal from the mesolimbic dopamine pathway (Chabrat et al., 2017)
<i>GPNMB</i> $\uparrow$ 7p15.3 SHAP rank 3	Transmembrane glycoprotein NMB Transmembrane glycoprotein, annotated as a possible melanogenic enzyme. (The UniProt Consortium, 2024) [Q14956]	effect  $g +0.97, q 0.003$ folds  92% of folds bulk  $g +0.43, q 0.070$ kept  $g +0.53, q 0.086$	-0.5 SOX6 AGTR1	0.503 genetic 0.782	<i>antibody:</i> advanced clinical 1 drug or clinical candidate	Human GWAS: risk locus, the risk allele raising GPNMB; cell and mouse models: GPNMB binds α -synuclein and promotes its neuronal uptake (Diaz-Ortiz et al., 2022) Plasma: raised in Parkinson’s disease across 731 patient and 59 control samples, tracking severity (Diaz-Ortiz et al., 2022)

Table 3, continued

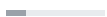
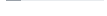
Gene	Type and main role	This study	DA-neuron subtype	Open Targets	Druggability	Published link to Parkinson's disease or dopamine neurons
<i>SCAI</i> ↓ 9q33.3 SHAP rank 4	Protein SCAI Suppressor of MRTF-A-driven SRF transcription that regulates cell migration. (The UniProt Consortium, 2024) [Q8N9R8]	effect  $g -0.66, q 0.003$ folds  72% of folds bulk  $g +0.13, q 0.699$ kept  $g -0.61, q 0.061$	-0.2 CALB1 PPP1R17	0.000	<i>antibody</i> : signal peptide or transmembrane	
<i>ICE1</i> ↑ 5p15.32 SHAP rank 5	Little elongation complex subunit 1 Scaffold of the little elongation complex, which drives small nuclear RNA transcription. (The UniProt Consortium, 2024) [Q9Y2F5]	effect  $g +0.73, q 0.017$ folds  80% of folds bulk  $g +0.14, q 0.699$ kept  $g +0.69, q 0.053$	+0.0	0.000		
<i>ZMYND8</i> ↑ 20q13.12 SHAP rank 6	MYND-type zinc finger-containing chromatin reader ZMYND8 Chromatin reader that recognises combined histone methylation and acetylation marks. (The UniProt Consortium, 2024) [Q9ULU4]	effect  $g +0.89, q 0.003$ folds  84% of folds bulk  $g +0.29, q 0.258$ kept  $g +0.66, q 0.053$	-1.6 SOX6 GFRA2	0.004		
<i>CNTN4</i> ↓ 3p26.3-p26.2 SHAP rank 7	Contactin-4 Cell-surface adhesion molecule of nervous-system development that promotes neurite outgrowth. (The UniProt Consortium, 2024) [Q8IWW2]	effect  $g -0.63, q 0.005$ folds  72% of folds bulk  $g +0.10, q 0.752$ kept  $g -0.16, q 0.539$	-3.1 SOX6 AGTR1	0.011	<i>antibody</i> : surface/secreted (high confidence)	Patient iPSC dopamine neurons, two brothers with the same PRKN mutations: CNTN4 was among the most down-regulated genes in the more severely affected brother (Cukier et al., 2022) Cerebrospinal fluid: measured across dementias, including Parkinson's disease dementia and dementia with Lewy bodies (Muqaku et al., 2024)
<i>CFLAR</i> ↑ 2q33.1 SHAP rank 8	CASP8 and FADD-like apoptosis regulator Apoptosis regulator that inhibits death-receptor-mediated cell death. (The UniProt Consortium, 2024) [O15519]	effect  $g +0.54, q 0.009$ folds  56% of folds bulk  $g -0.11, q 0.752$ kept  $g +0.32, q 0.225$	+0.7 CALB1 GEM	0.005	<i>small molecule</i> : structure with ligand <i>antibody</i> : surface/secreted (HPA)	

Table 3, continued

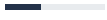
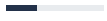
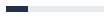
Gene	Type and main role	This study	DA-neuron subtype	Open Targets	Druggability	Published link to Parkinson’s disease or dopamine neurons
<i>PPFIA4</i> ↑ 1q32.1 SHAP rank 9	Liprin-alpha-4 Scaffold that may regulate focal-adhesion turnover and position receptor tyrosine phosphatases. (The UniProt Consortium, 2024) [O75335] excitatory synapse	effect  $g +0.82, q 0.003$ folds  76% of folds bulk  $g +0.46, q 0.057$ kept  $g +0.46, q 0.122$	+1.0 CALB1 PPP1R17	0.007	<i>antibody</i> : surface/secreted (medium confidence)	
<i>CALB1</i> ↑ 8q21.3 SHAP rank 10	Calbindin Calcium-binding protein that buffers cytosolic calcium. (The UniProt Consortium, 2024) [P05937] presynaptic Ca²⁺ binding	effect  $g +0.57, q 0.007$ folds  52% of folds bulk  $g +0.45, q 0.057$ kept  $g +0.24, q 0.355$	+15.4 CALB1 CALCR	0.084		Human post-mortem nigra: calbindin-rich compartments lose fewer dopamine neurons (Damier et al., 1999)
<i>DNAJB1</i> ↑ 19p13.12 SHAP rank 11	DnaJ homolog subfamily B member 1 HSP40 co-chaperone that stimulates HSP70 in protein folding. (The UniProt Consortium, 2024) [P25685]	effect  $g +0.68, q 0.011$ folds  56% of folds bulk  $g +0.37, q 0.129$ kept  $g +0.53, q 0.086$	-3.0 SOX6 DDT	0.089	<i>small molecule</i> : structure with ligand	Cell model: DNAJB1 clears α -synuclein through the Hsp70 system, and its phosphorylation by EGFR suppresses aggregation (Huang et al., 2025) In vitro: DNAJB1 binds α -synuclein fibrils and recruits Hsc70 to take them apart (Monistrol et al., 2025)
<i>CALB2</i> ↑ 16q22.2 SHAP rank 12	Calretinin Calcium-binding protein that buffers intracellular calcium in specific neuronal populations. (The UniProt Consortium, 2024) [P22676] presynaptic Ca²⁺ binding, excitatory synapse	effect  $g +1.04, q 0.003$ folds  52% of folds bulk  $g +0.35, q 0.135$ kept  $g +0.33, q 0.224$	+3.7 CALB1 GEM	0.038		Human post-mortem midbrain, 4 patients and 3 controls: calretinin-positive dopamine neurons of the substantia nigra were selectively preserved (Mouatt-Prigent et al., 1994) Rat culture: calretinin-containing dopamine neurons were spared from levodopa toxicity (Isaacs et al., 1997) Plasma: among the proteins most reduced in Parkinson’s disease compared with controls, by nucleic acid-linked immunoassay (Vijaratnam et al., 2026)
<i>SH3GL3</i> ↓ 15q25.2 SHAP rank 13	Endophilin-A3 Endophilin that binds curved membranes and takes part in endocytosis. (The UniProt Consortium, 2024) [Q99963]	effect  $g -0.69, q 0.014$ folds  52% of folds bulk  $g -0.15, q 0.699$ kept  $g -0.66, q 0.053$	-4.0 SOX6 AGTR1	0.002	<i>antibody</i> : surface/secreted (medium confidence)	

Table 3, continued

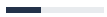
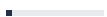
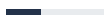
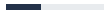
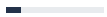
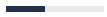
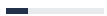
Gene	Type and main role	This study	DA-neuron subtype	Open Targets	Druggability	Published link to Parkinson's disease or dopamine neurons
<i>GSTK1</i> ↑ 7q34 SHAP rank 14	Glutathione S-transferase kappa 1 Glutathione S-transferase that conjugates glutathione to foreign and endogenous compounds. (The UniProt Consortium, 2024) [Q9Y2Q3]	effect  $g +0.62, q 0.005$ folds  36% of folds bulk  $g +0.04, q 0.821$ kept  $g +0.55, q 0.083$	+0.5 SOX6 DDT	0.003	<i>small molecule</i> : structure with ligand	
<i>ADGRB3</i> ↓ 6q12-q13 SHAP rank 15	Adhesion G protein-coupled receptor B3 Adhesion G protein-coupled receptor that regulates synapse and dendritic-spine formation. (The UniProt Consortium, 2024) [O60242] excitatory synapse, climbing-fibre synapse, synapse pruning	effect  $g -0.52, q 0.011$ folds  36% of folds bulk  $g +0.00, q 0.993$ kept  $g -0.36, q 0.191$	+0.0	0.000	<i>small molecule</i> : structure with ligand <i>antibody</i> : surface/secreted (high confidence)	
<i>L3MBTL1</i> ↑ 20q13.12 SHAP rank 16	Lethal(3)malignant brain tumor-like protein 1 Polycomb-group reader of methylated lysines that keeps genes repressed. (The UniProt Consortium, 2024) [Q9Y468]	effect  $g +0.44, q 0.077$ folds  16% of folds bulk  $g +0.10, q 0.752$ kept  $g +0.50, q 0.099$	+0.0 CALB1 GEM	0.000	<i>small molecule</i> : structure with ligand	
<i>C11orf54</i> ↓ nan SHAP rank 17	Beta-keto L-gulonate decarboxylase Decarboxylase of the pentose pathway that also regulates HIF1A degradation. (The UniProt Consortium, 2024) [Q9H0W9]	effect  $g -0.71, q 0.007$ folds  40% of folds bulk  $g -0.15, q 0.725$ kept  $g -0.68, q 0.053$	-0.7 SOX6 GFRA2	0.000		
<i>IBTK</i> ↓ 6q14.1 SHAP rank 18	Inhibitor of Bruton tyrosine kinase Inhibitor of Bruton tyrosine kinase in B-cell development. (The UniProt Consortium, 2024) [Q9P2D0]	effect  $g -0.90, q 0.003$ folds  60% of folds bulk  $g -0.13, q 0.699$ kept  $g -0.76, q 0.053$	-1.4 SOX6 AGTR1	0.000	<i>antibody</i> : surface/secreted (medium confidence)	

Table 3, continued

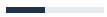
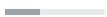
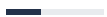
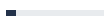
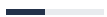
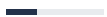
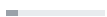
Gene	Type and main role	This study	DA-neuron subtype	Open Targets	Druggability	Published link to Parkinson's disease or dopamine neurons
<i>CPNE3</i> ↓ 8q21.3 SHAP rank 19	Copine-3 Calcium-dependent phospholipid-binding protein involved in cell migration. (The UniProt Consortium, 2024) [O75131]	effect  g -0.56, q 0.030 folds  40% of folds bulk  g +0.23, q 0.461 kept  g -0.36, q 0.191	-8.9 SOX6 AGTR1	0.000	<i>antibody</i> : surface/secreted (high confidence)	
<i>FUS</i> ↑ 16p11.2 SHAP rank 20	RNA-binding protein FUS DNA/RNA-binding protein acting in transcription, splicing, RNA transport and DNA repair. (The UniProt Consortium, 2024) [P35637]	effect  g +0.89, q 0.005 folds  36% of folds bulk  g +0.07, q 0.752 kept  g +0.84, q 0.042	+0.0	0.031	<i>small molecule</i> : structure with ligand	
<i>ELOVL7</i> ↓ 5q12.1 SHAP rank 21	Very long chain fatty acid elongase 7 Rate-limiting elongase of long- and very-long-chain fatty acids in the endoplasmic reticulum. (The UniProt Consortium, 2024) [A1L3X0]	effect  g -0.68, q 0.007 folds  40% of folds bulk  g -0.29, q 0.461 kept  g -0.52, q 0.088	-3.1 SOX6 AGTR1	0.316 genetic 0.507	<i>small molecule</i> : structure with ligand <i>antibody</i> : signal peptide or transmembrane	Human case-control study: variant associated with Parkinson's disease (Li et al., 2018)
<i>PRKAR2A</i> ↑ 3p21.31 SHAP rank 22	cAMP-dependent protein kinase type II-alpha regulatory subunit Regulatory subunit of cAMP-dependent protein kinase that anchors it to membranes. (The UniProt Consortium, 2024) [P13861]	effect  g +0.69, q 0.021 folds  32% of folds bulk  g -0.08, q 0.752 kept  g +0.63, q 0.053	-0.0 SOX6 GFRA2	0.020	<i>small molecule</i> : structure with ligand <i>antibody</i> : surface/secreted (high confidence)	Human Lewy body disease brain and mice: α -synuclein sequesters DNMT1 from the nucleus, hypomethylating CpG islands upstream of PRKAR2A (Desplats et al., 2011)
<i>USP12</i> ↓ 13q12.13 SHAP rank 23	Ubiquitin carboxyl-terminal hydrolase 12 Deubiquitinating enzyme that regulates immune and inflammatory signalling. (The UniProt Consortium, 2024) [O75317]	effect  g -1.06, q 0.003 folds  48% of folds bulk  g -0.21, q 0.480 kept  g -0.60, q 0.061	-4.9 SOX6 AGTR1	0.023	<i>small molecule</i> : structure with ligand <i>antibody</i> : surface/secreted (high confidence)	

Table 3, continued

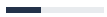
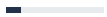
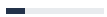
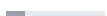
Gene	Type and main role	This study	DA-neuron subtype	Open Targets	Druggability	Published link to Parkinson's disease or dopamine neurons
<i>SOX6</i> ↓ 11p15.2 SHAP rank 24	Transcription factor SOX-6 Transcription factor in neurogenesis and cartilage development. (The UniProt Consortium, 2024) [P35712]	effect  $g -0.77, q 0.004$ folds  36% of folds bulk  $g -0.41, q 0.210$ kept  $g -0.33, q 0.224$	-18.3 SOX6 AGTR1	0.087		Human single-nucleus nigra: marks the dopamine population depleted in Parkinson's disease (Kamath et al., 2022) Mouse: legumain cleaves SOX6 in vulnerable dopamine neurons, and the fragment speeds degeneration in A53T α -synuclein mice (Nie et al., 2024)
<i>MANF</i> ↑ 3p21.2 SHAP rank 25	Mesencephalic astrocyte-derived neurotrophic factor Secreted factor that selectively promotes survival of ventral midbrain dopamine neurons. (The UniProt Consortium, 2024) [P55145]	effect  $g +0.55, q 0.020$ folds  16% of folds bulk  $g +0.39, q 0.103$ kept  $g +0.37, q 0.191$	-1.0 SOX6 DDT	0.102	<i>antibody:</i> surface/secreted (high confidence)	Rat 6-hydroxydopamine model: MANF restores nigrostriatal function (Voutilainen et al., 2009)
<i>LGMN</i> ↓ 14q32.12 SHAP rank 26	Legumain Lysosomal protease that cleaves proteins after asparagine residues. (The UniProt Consortium, 2024) [Q99538]	effect  $g -0.46, q 0.031$ folds  20% of folds bulk  $g +0.13, q 0.699$ kept  $g -0.24, q 0.355$	+0.0	0.082	<i>small molecule:</i> structure with ligand <i>antibody:</i> surface/secreted (high confidence)	Cell, mouse and human brain: legumain cleaves α -synuclein at N103, and the fragment aggregates and is toxic to dopamine neurons (Zhang et al., 2017) Mouse: it cleaves α -synuclein and tau and drives their spread from gut to brain (Ahn et al., 2019)
<i>SSTR1</i> ↓ 14q21.1 SHAP rank 27	Somatostatin receptor type 1 Somatostatin receptor that inhibits adenylyl cyclase through G proteins. (The UniProt Consortium, 2024) [P30872]	effect  $g -0.93, q 0.003$ folds  48% of folds bulk  $g -0.07, q 0.752$ kept  $g -0.44, q 0.136$	-6.8 SOX6 DDT	0.075	<i>small molecule:</i> structure with ligand <i>antibody:</i> surface/secreted (high confidence) 8 drug or clinical candidates	
<i>BCL11A</i> ↑ 2p16.1 SHAP rank 28	BCL11 transcription factor A Transcription factor associated with the BAF chromatin-remodelling complex. (The UniProt Consortium, 2024) [Q9H165]	effect  $g +0.70, q 0.003$ folds  40% of folds bulk  $g +0.16, q 0.699$ kept  $g +0.40, q 0.161$	+0.7 SOX6 GFRA2	0.008	1 drug or clinical candidate	Mouse: marks a subset of nigral dopamine neurons that is especially vulnerable to α -synuclein overexpression and oxidative stress (Tolve et al., 2021)

Table 3, continued

Gene	Type and main role	This study	DA-neuron subtype	Open Targets	Druggability	Published link to Parkinson’s disease or dopamine neurons
<i>CACNA1A</i> ↑ 19p13.13 SHAP rank 29	Voltage-dependent P/Q-type calcium channel subunit alpha-1A Pore-forming subunit of P/Q-type voltage-gated calcium channels. (The UniProt Consortium, 2024) [O00555]	effect  $g +0.48, q 0.068$ folds  24% of folds bulk  $g +0.08, q 0.752$ kept  $g +0.55, q 0.083$	-2.8 SOX6 AGTR1	0.355	<i>small molecule</i> : approved drug <i>antibody</i> : surface/secreted (high confidence) 11 drug or clinical candidates	Cell model: the Parkinson’s kinase LRRK2 regulates the Cav2.1 channel (Bedford et al., 2016)
<i>TRRAP</i> ↑ 7q22.1 SHAP rank 30	Transformation/transcription domain-associated protein Adapter of histone acetyltransferase complexes that mark chromatin for transcription. (The UniProt Consortium, 2024) [Q9Y4A5]	effect  $g +0.83, q 0.004$ folds  32% of folds bulk  $g +0.06, q 0.772$ kept  $g +0.65, q 0.053$	+0.6 CALB1 CALCR	0.333	<i>small molecule</i> : structure with ligand	